

Homework 1

Name:

Problem 1

Bisection Method

Given the function

$$f(\theta) = \sin(\theta) - \theta^2$$

complete the following:

- (a) Plot the function over an appropriate range of θ values using matplotlib.
- (b) Determine initial bracketing values for each root from the plot.
- (c) Implement the bisection method in Python and compute the values of each root to within 10^{-8} .

Problem 2

Comparing Bracketing Methods

- (a) How many iterations of the bisection method are needed to approximate the root of

$$\cos(\theta) = \theta$$

on the interval $\theta \in [-1, 1]$ to within 10^{-4} ?

- (b) Implement the Regula Falsi method in Python and use it to solve the same problem. How many iterations does it take to compute the root to the same accuracy?
- (c) Using matplotlib, create a plot to compare the error of the methods above. Plot the relative error on the y -axis against iteration count on the x -axis.

Problem 3

Fixed-Point Iteration

- (a) Implement fixed-point iteration to locate the root $x = 0$ of

$$f(x) = \sin(\sqrt{x}) - x.$$

Use an initial guess of $x_0 = 0.5$ and iterate until your result is within 10^{-4} of the true solution.

- (b) Plot the relative error and the true error against the iteration count. Make sure your plot includes a legend so that it's easy to tell which is which.

Problem 4**Newton-Raphson Iteration**

Determine the real roots of the following polynomials:

$$f_1(x) = -0.87x^2 + 1.65x + 8.25$$

$$f_2(x) = -1.9 + 6.31x - 3.7x^2 + 0.7x^3$$

- (a) Plot each function and estimate the real roots graphically.
- (b) Implement Newton-Raphson iteration to find the root to within 10^{-8} .
- (c) Finally, plug your solution back into the function to check your result.

Problem 5**Bracketing vs. Open Methods**

Apply the bisection method, Regula Falsi, Newton-Raphson iteration, and the Secant method to find the positive solution to

$$3 \sin(\theta) + \cos(\theta) = \theta^3 - 9$$

to within 10^{-4} .

Create a plot showing the value produced by each numerical method at each iteration. Make sure to use a legend to indicate which values came from each method.

Problem 6**Optimization Methods**

Use the following methods with the given values to find the maxima of the function

$$f(x) = -0.3x^4 + 1.8x^3 - 1.2x^2 + 2x$$

- (a) Golden-section search ($x_l = -2$, $x_u = 4$, $\varepsilon_s = 1\%$)
- (b) Parabolic interpolation ($x_1 = 1.7$, $x_2 = 2$, $x_3 = 2.7$, 5 iterations)